

Wireless PIR Motion Detector

Passive infrared presence sensor — 1 or 2 elements — 433.92 MHz RF transmitter with EETX433A

Application guide for wireless presence alarms using the EETX433A multifunction transmitter in passive sensor (PIR) mode, in single-element and dual-element (AND-gate) variants.

FREQUENCY

433.92 MHz

RF POWER

13 dBm

VARIANTS

1 PIR · 2 PIR

RANGE

10–12 m

VERSION

v1.0

BASED ON IC

EETX433A

DEVELOPED BY

Engeletron Engenharia

DATE

2026-06-25

• Introduction

This application note describes how to build a **wireless motion detector** based on a passive infrared (PIR) sensor using the **EETX433A** multifunction transmitter in passive sensor mode. The EETX433A integrates a 433.92 MHz PLL oscillator, OOK/ASK modulator, power amplifier and the encoding and power-management logic in a single SOP-8 package, enabling a compact, ultra-low-power battery-powered product.

Two variants are presented that share the same firmware and the same RF transmission format, differing only in the number of PIR elements and in the trigger logic: the **1 PIR** variant fires on each motion detection of a single sensor; the **2 PIR** variant adds an ultra-low-power **AND** logic gate and only fires when both sensors detect motion **simultaneously**, reducing false alarms.

The IC electrical details (pinout, absolute ratings, operating conditions, RF frame format and code mapping) are given in the EETX433A datasheet and referenced throughout this document.

• The Two Variants

1 PIR variant (single element)

A digital pyrosensor with ON/OFF output is wired directly to the sensor input (SENIN) of the EETX433A. On each detected motion, the sensor's rising edge triggers an RF transmission with the violation code. This is the simplest, most economical configuration, suited to common indoor environments.

2 PIR variant (dual-lens with AND gate)

Two independent digital pyrosensors have their outputs combined by an ultra-low-power **AND** logic gate; only the gate output goes to the EETX433A input. Thus the RF transmission only occurs when **both** sensors detect motion at the same time. This coincidence drastically reduces false triggers caused by air currents, insects or localized thermal changes, while keeping the same transmitted code as the 1 PIR variant.

- **The Two Variants**

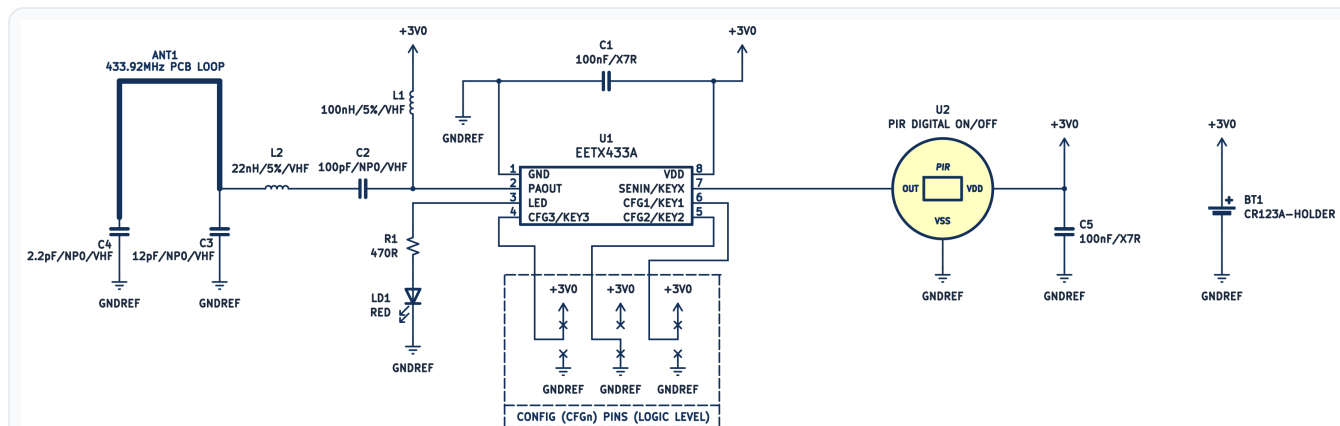


Figure 1: 1 PIR variant — single digital pyrosensor driving the EETX433A

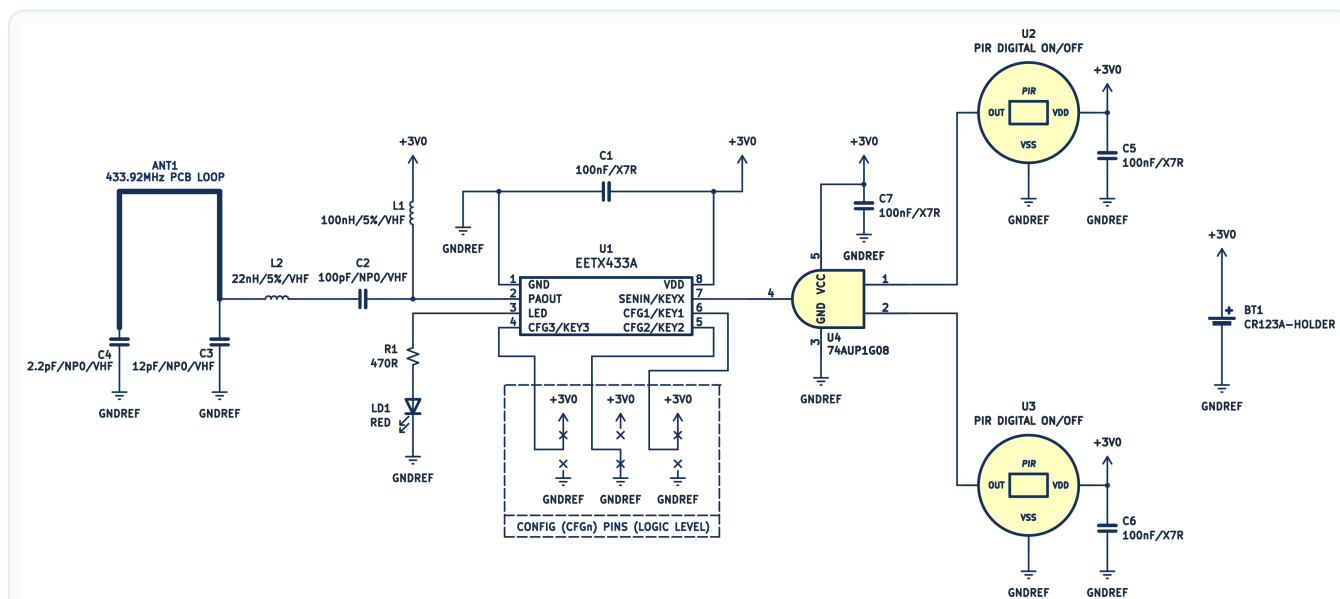


Figure 2: 2 PIR variant — two pyrosensors + AND gate (coincidence trigger) + EETX433A

• Operating Sequence

After the battery is inserted, the sensor goes through four phases. The LED visually indicates each stage. The times below are fixed in firmware.

Phase	Duration	Re-arm	LED	Behavior
Initialization	2 s	—	On for 2 s	Visually confirms the sensor powered up and initialized correctly.
Stabilization	5 s	Restarts on any motion	Off	Environment reading. Do not move in front of the sensor for 5 continuous seconds; any motion restarts the count.
Test mode	10 min	5 s without motion	Fast blink 1.5 s on detection	Detection with fast re-arm to ease installation. Each detection sends the violation code.
Normal mode	Continuous	120 s without motion	Fast blink on detection	Low-power operation. After each trigger, 2 min without motion are required to re-arm.

RE-ARM RULE

To re-arm after a detection, a continuous interval without any motion is required (5 s in Test mode, 120 s in Normal mode). If motion occurs during that interval, the timer restarts and extends the pause until the motion-free time is met. Only then is the sensor ready for a new detection and transmission.

• Sensitivity

The sensor features **automatic sensitivity with temperature compensation**. The gain is self-adaptive according to the environment and the average amount of motion, requiring no manual adjustment (no sensitivity trimpot or jumper). This keeps detection stable in both cold and warm environments.

• Low-Battery Detection

The EETX433A continuously monitors the supply voltage in sensor mode. The nominal low-battery threshold is **2.20 V**, with $\pm 5\%$ (± 0.10 V) tolerance — in practice the signaling may occur between 2.30 V and 2.10 V depending on component variation.

To avoid false warnings from momentary voltage dips, the condition must be confirmed on **two consecutive trigger events** with low voltage; only from the second event does the sensor signal low battery.

Once low battery is confirmed, on each motion detection the LED blinks **3 times quickly**, waits **1 s** and transmits the **violation code**; it then waits another **1 s** and transmits the **low-battery code**. The receiver can then alert the user to replace the battery.

• Transmitted RF Signal

Transmission follows the EETX433A packet format: pulse-width OOK encoding at 433.92 MHz, with a basic time unit $\lambda = 500 \mu\text{s}$ (each bit takes $3\lambda = 1.5$ ms). Each packet contains pilot, start, 20 bits of programmable address (factory-unique), 4 bits of data and 4 bits of fixed anti-code, totaling $108\lambda \approx 54$ ms.

In passive sensor mode, the 4-bit data field carries the event: the **violation code** on each motion detection and the **low-battery code** when the undervoltage condition is confirmed. The 20-bit address uniquely identifies the sensor at the receiver. See the EETX433A datasheet for the full frame structure and code mapping.

• Technical Characteristics

Values for the complete sensor system (EETX433A IC + PIR + battery). Rest current differs between variants due to the number of active pyrosensors.

Parameter	1 PIR	2 PIR
Operating frequency	433.92 MHz	433.92 MHz
Channels	1	1
Modulation	OOK / ASK	OOK / ASK
RF power (typ.)	13 dBm	13 dBm
Supply voltage	3.0 V nom. (2.0–3.6 V)	3.0 V nom. (2.0–3.6 V)
Battery options	1× CR2032 / 1× CR123 / 2× AA(AAA)	1× CR2032 / 1× CR123 / 2× AA(AAA)
Transmit current	12 mA	12 mA
Rest current	12 µA	22 µA
Detection range	10–12 m	10–12 m
Trigger logic	Motion (1 sensor)	AND coincidence (2 sensors)
Low-battery threshold	2.20 V ±5%	2.20 V ±5%
IC package	SOP-8	SOP-8

IC RF and standby per the EETX433A datasheet (13 dBm; 800 nA standby). System rest current is dominated by the pyrosensor(s).

• Application Guidelines

- **Placement:** install the sensor at 2.0–2.2 m height, with the lens facing the area to protect and the expected motion **crossing** the field of view (higher sensitivity to transverse than to radial motion).
- **Avoid heat sources:** keep the sensor away from direct sunlight, headlights, heating/air-conditioning units and surfaces that change temperature quickly.
- **Initialization:** after inserting the battery, keep the area free of motion during stabilization (5 s) for a clean environment reading.
- **Antenna:** use the printed 433.92 MHz loop antenna per the reference schematic and the EETX433A datasheet; avoid routing ground plane or nearby metal around the antenna.
- **2 PIR variant:** align both elements so they cover the same crossing zone — the trigger only occurs on the coincidence of both detections.

• Soldering and Reflow

Excess temperature can damage the EETX433A, causing thermal deformation of the package and a shift of the 433.92 MHz frequency. The reflow limit is **$\leq 210\text{ °C}$ for $\leq 10\text{ s}$** — the component's own maximum rating, more restrictive than the standard J-STD-020 reflow classifications (220–235 °C for Sn-Pb).

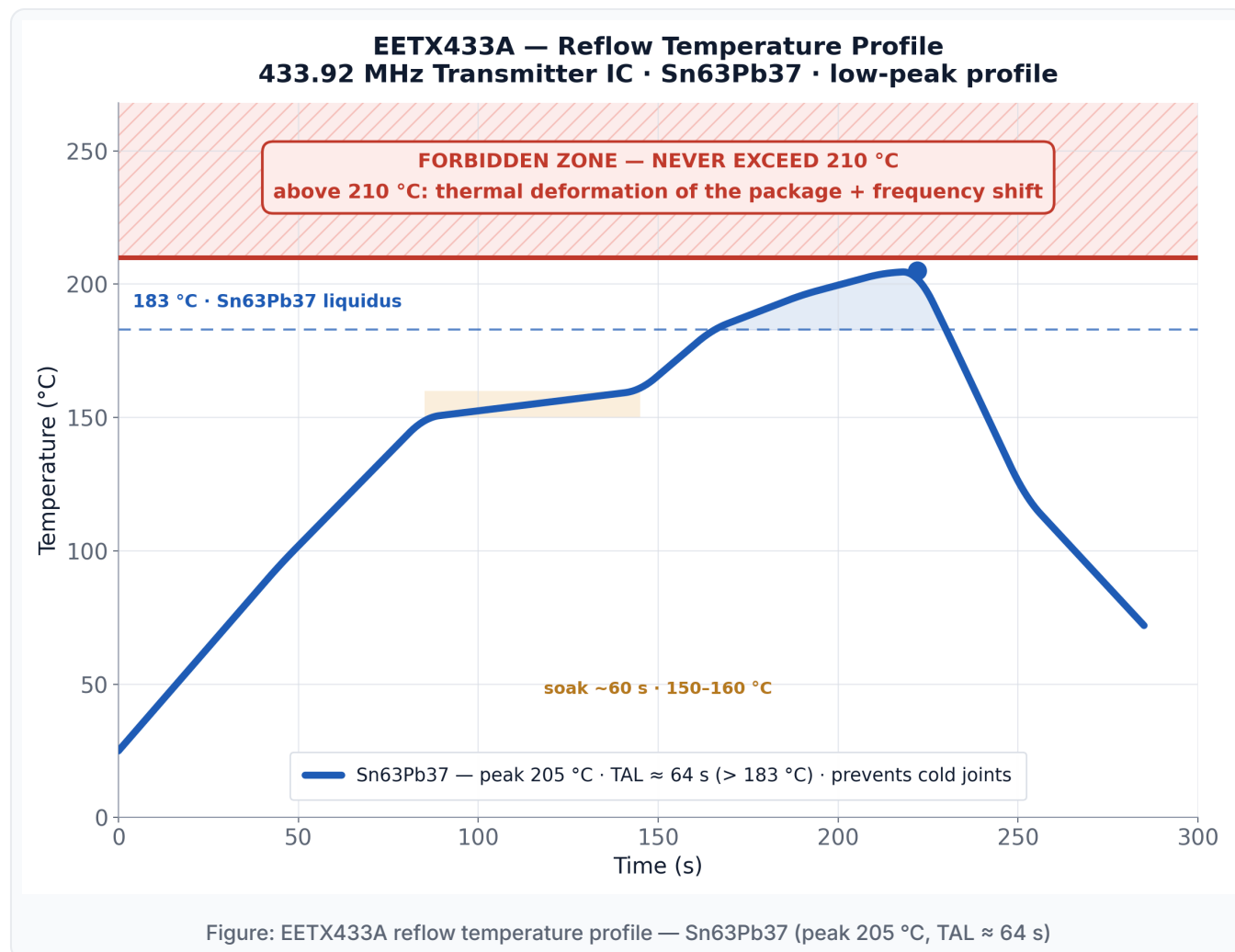
Alloy: **Sn63Pb37** (eutectic, 183 °C liquidus — the market standard), whose reflow peaks typically at **205–220 °C**; for the EETX433A the peak stays at the low end: board peak **205–210 °C** (target ~205 °C, never above 210 °C), compensated by a **TAL of 60–75 s**. Because the peak is modest, the time above liquidus is extended — and it is this combination that forms the intermetallic without exceeding the component limit.

To avoid **cold joints**: keep the soak of ~60 s between 150–160 °C (activates the flux and evens the board) and the **TAL of ~64 s above 183 °C** (ensures full melt and good wetting). Ramp-up $\leq 2.5\text{ °C/s}$ and cooling $\leq 4\text{ °C/s}$. A high, short peak is not needed: a low peak plus an adequate TAL produces the same joint, safely for the IC.

NARROW WINDOW — MEASUREMENT IS MANDATORY

The Sn63Pb37 window here is only ~5 °C wide: **below 205 °C** there is a risk of cold joints; **above 210 °C** the IC is damaged (SOP-8 deformation and frequency drift). The profile **must be measured with a thermocouple on the board**, at least once per line. NEVER exceed 210 °C.

• Soldering and Reflow



Phase	Time (s)	Temperature (°C)
Ramp-up	0–85	25 → 150
Preheat (soak)	85–145	150 → 160
Ramp to peak	145–212	160 → 204
Peak (reflow)	212–228	205 (max. 210)
TAL — time above liquidus	≈ 64	> 183
Cooling	228–285	205 → 72

Sn63Pb37 (eutectic, 183 °C liquidus) — market standard. Low-peak profile: peak 205–210 °C (never above 210 °C), compensated by a TAL of 60–75 s. A ~60 s soak at 150–160 °C and a ~64 s TAL prevent cold joints; ramp-up ~1–2.5 °C/s · cooling ≤ 4 °C/s · max. 2 reflow cycles. Hand soldering (samples): tip ≤ 300 °C, contact ≤ 5 s.

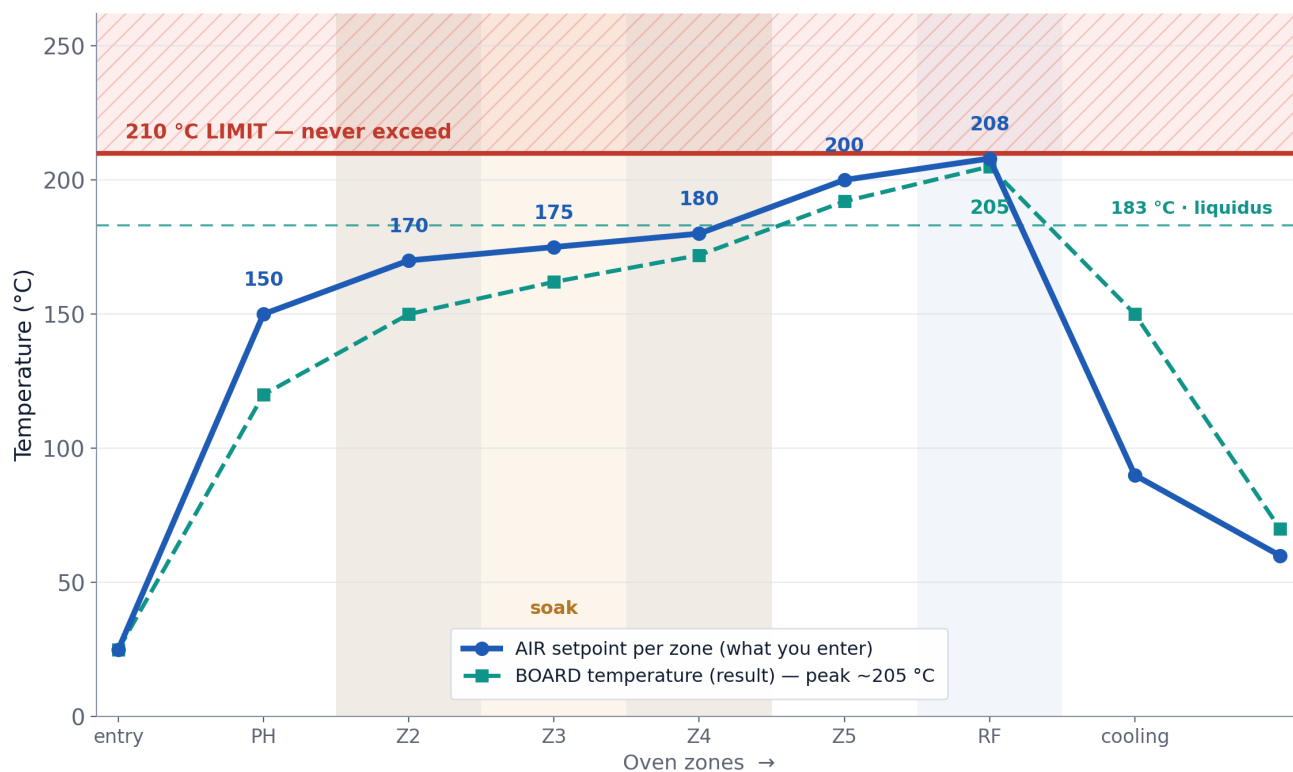
• Practical oven setup — zone profile

The chart below shows the reference setup for a tunnel oven: the **air setpoint per zone** (solid line — 150 · 170 · 175 · 180 · 200 · 208 °C) and the resulting **board temperature** (dashed). Air setpoints are a starting point — tune by profiling with a thermocouple on the board. What matters is the board result: **peak ~205 °C and TAL ~64 s**, never above 210 °C.

NARROW WINDOW — MEASUREMENT IS MANDATORY

The Sn63Pb37 window here is only ~5 °C wide: **below 205 °C** there is a risk of cold joints; **above 210 °C** the IC is damaged (SOP-8 deformation and frequency drift). The profile **must be measured with a thermocouple on the board**, at least once per line. NEVER exceed 210 °C.

EETX433A — Zone profile in straight lines
Sn63Pb37 · entered air setpoint × resulting board temperature



Zone profile — air setpoint × board temperature (peak ~205 °C · TAL ~64 s · ≤ 210 °C).

To learn more about the EETX433A and obtain the full datasheet, visit: engeletron.com.br

Legal Notice

The information in this application note is provided in good faith and believed to be accurate at the time of publication. Engletron Engenharia reserves the right to modify specifications without notice. The circuits and values shown are reference examples; it is the integrator's responsibility to validate performance in the final application. We accept no liability for consequences arising from the use of this information.